LAPORAN PRAKTIKUM

POST TEST 3

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

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Kelas B1 ‘25

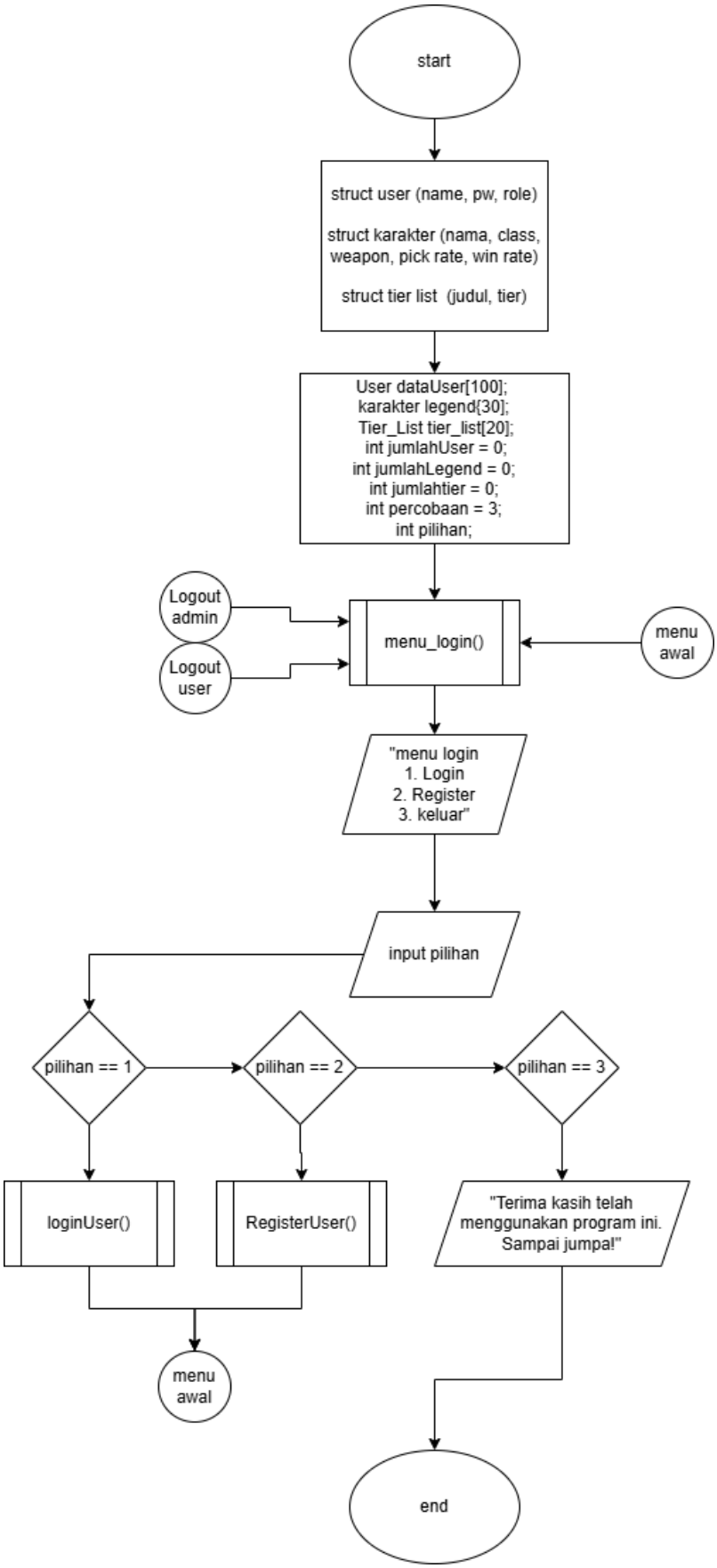
PROGRAM STUDI INFORMATIKA

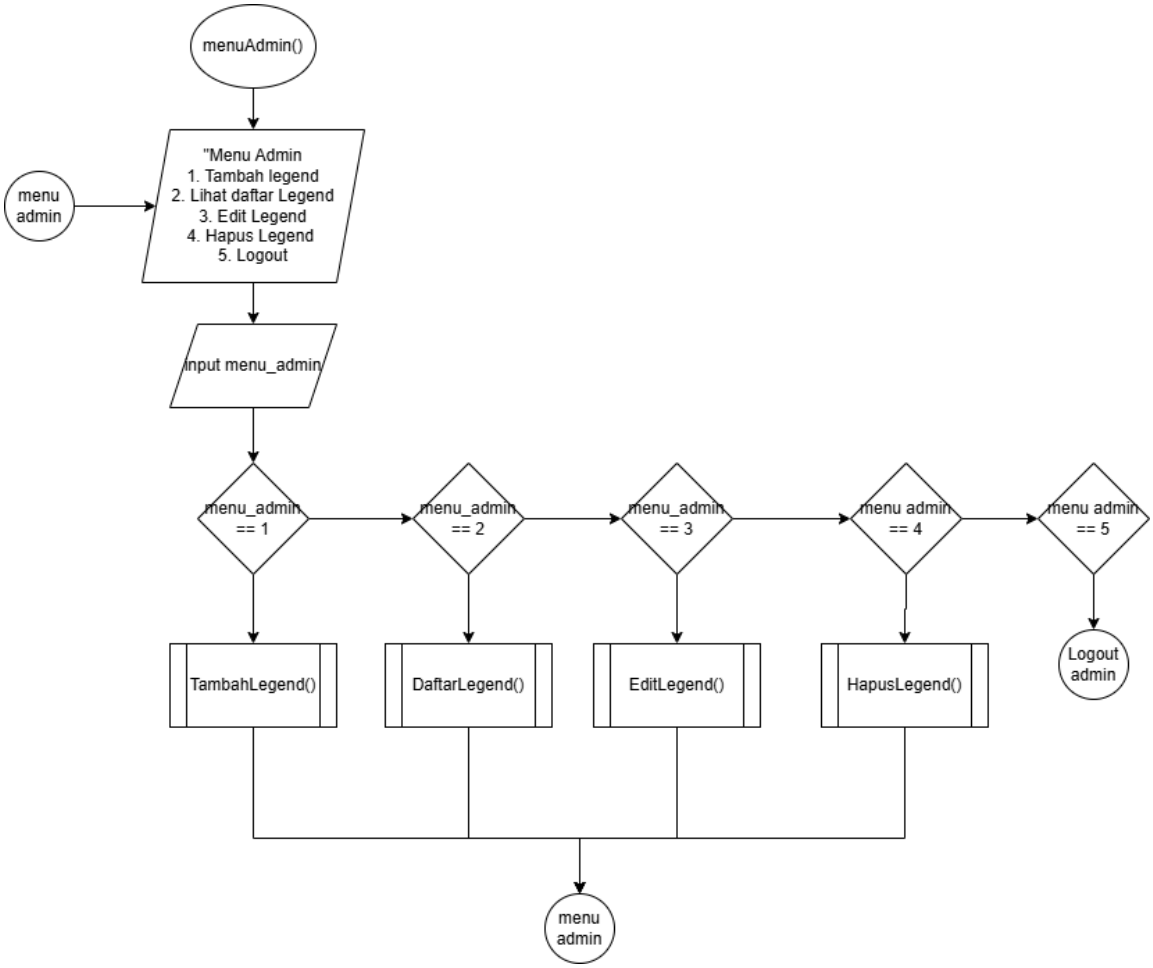
UNIVERSITAS MULAWARMAN

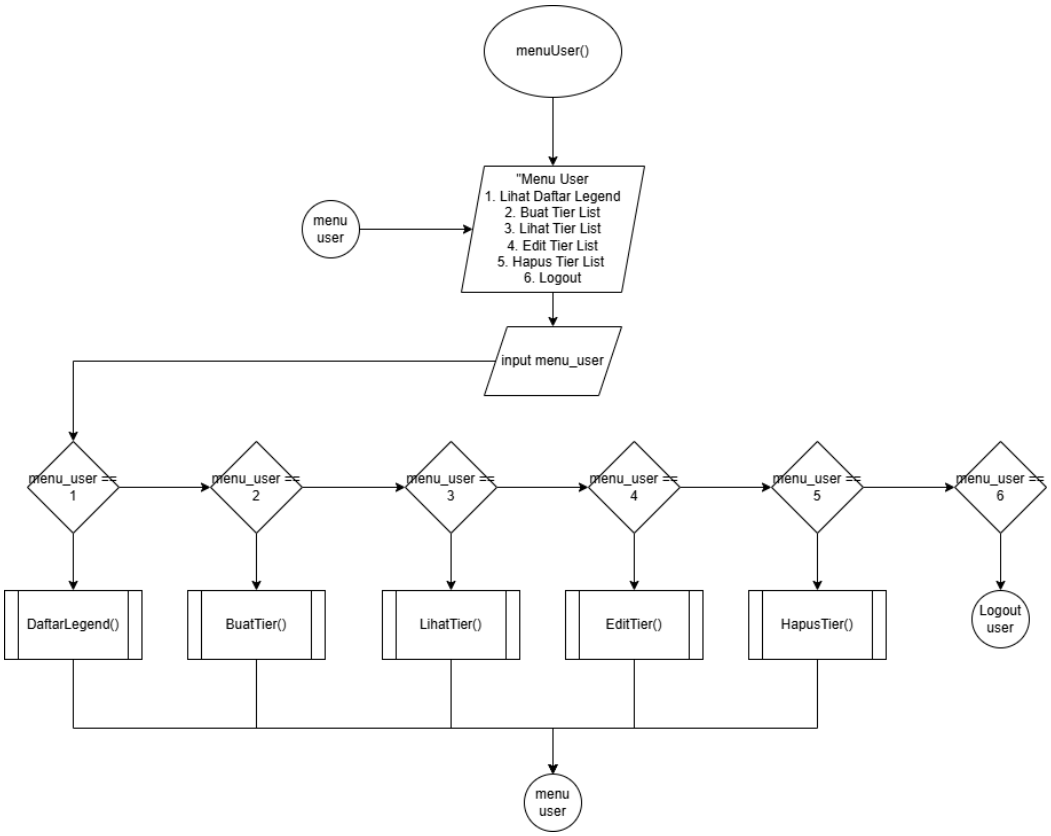
SAMARINDA

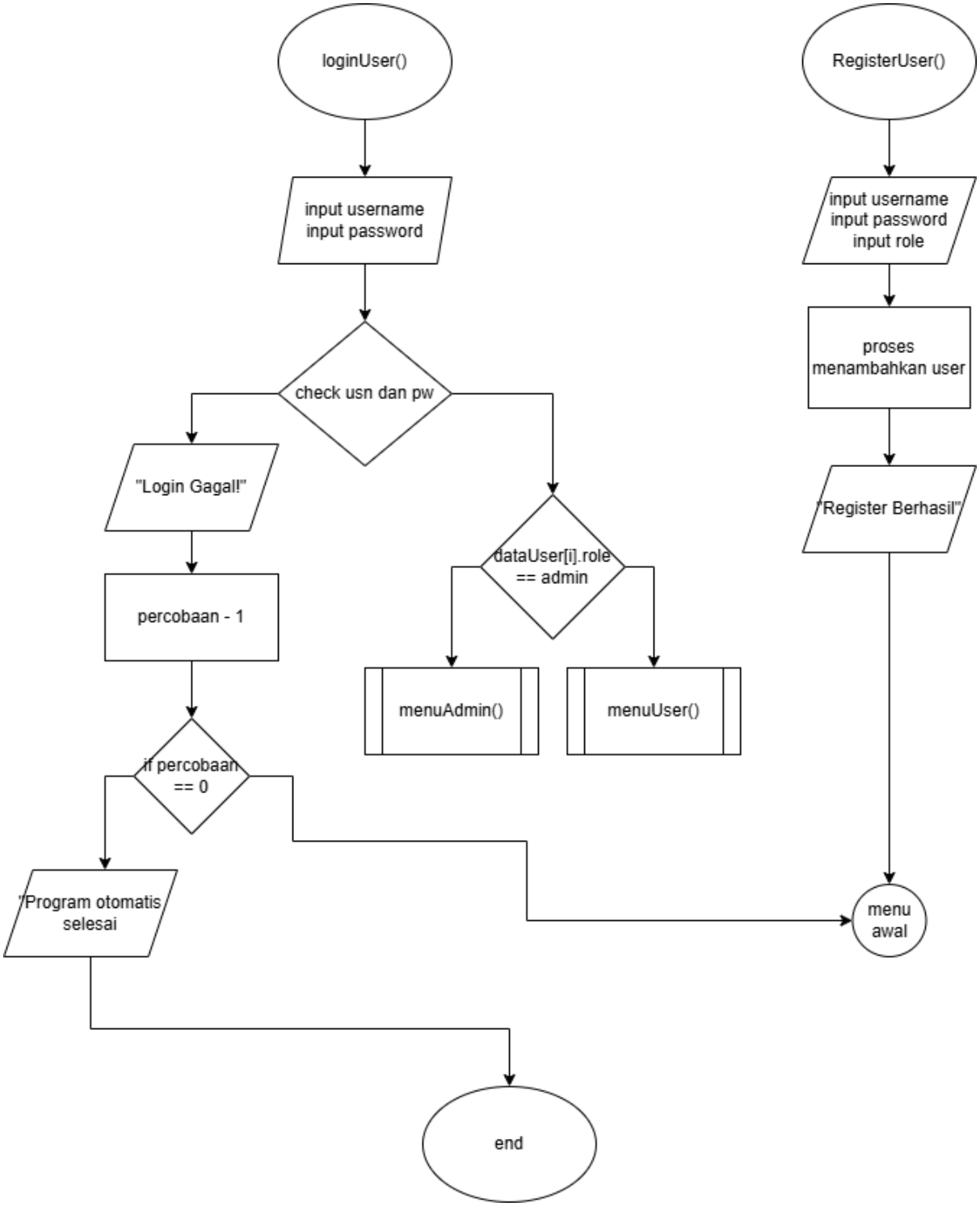
2025

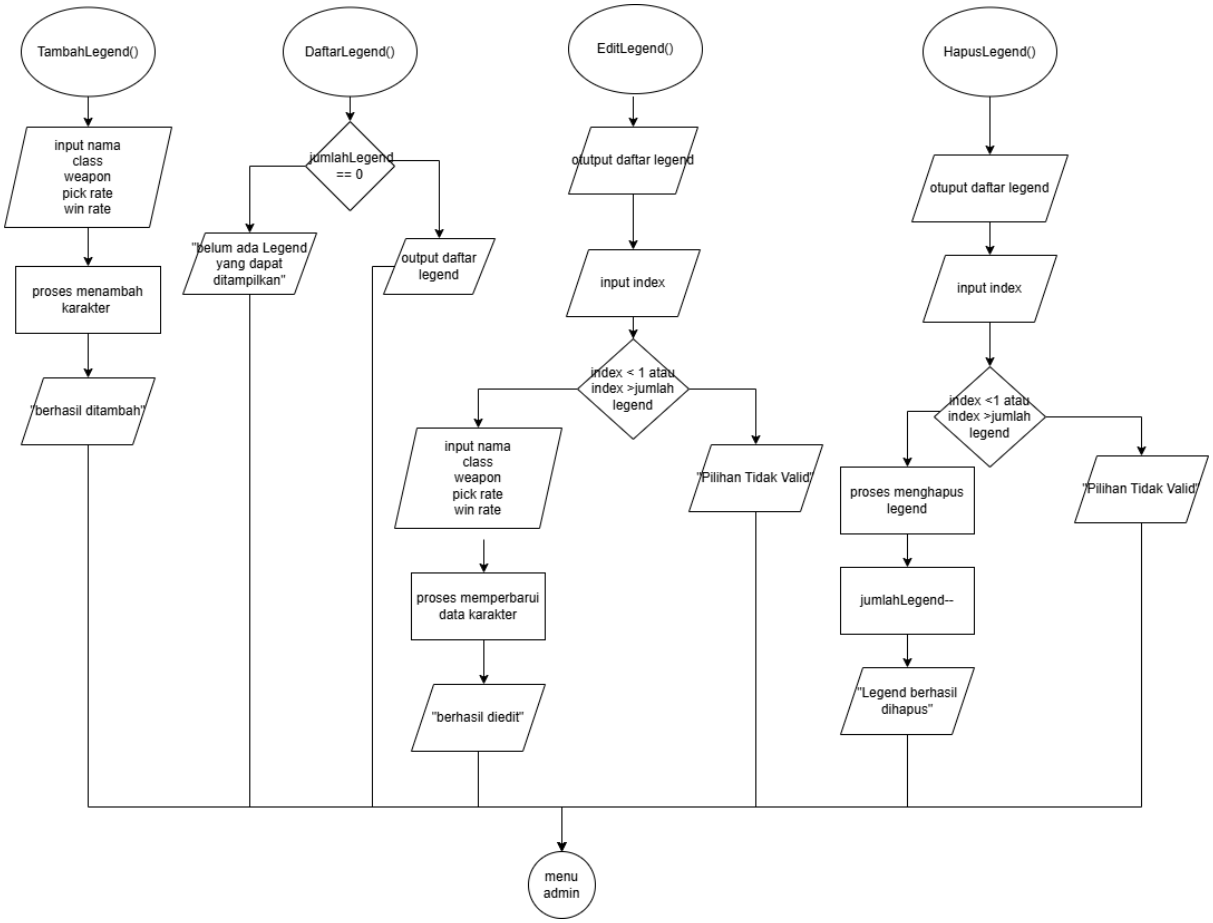
1. Flowchart

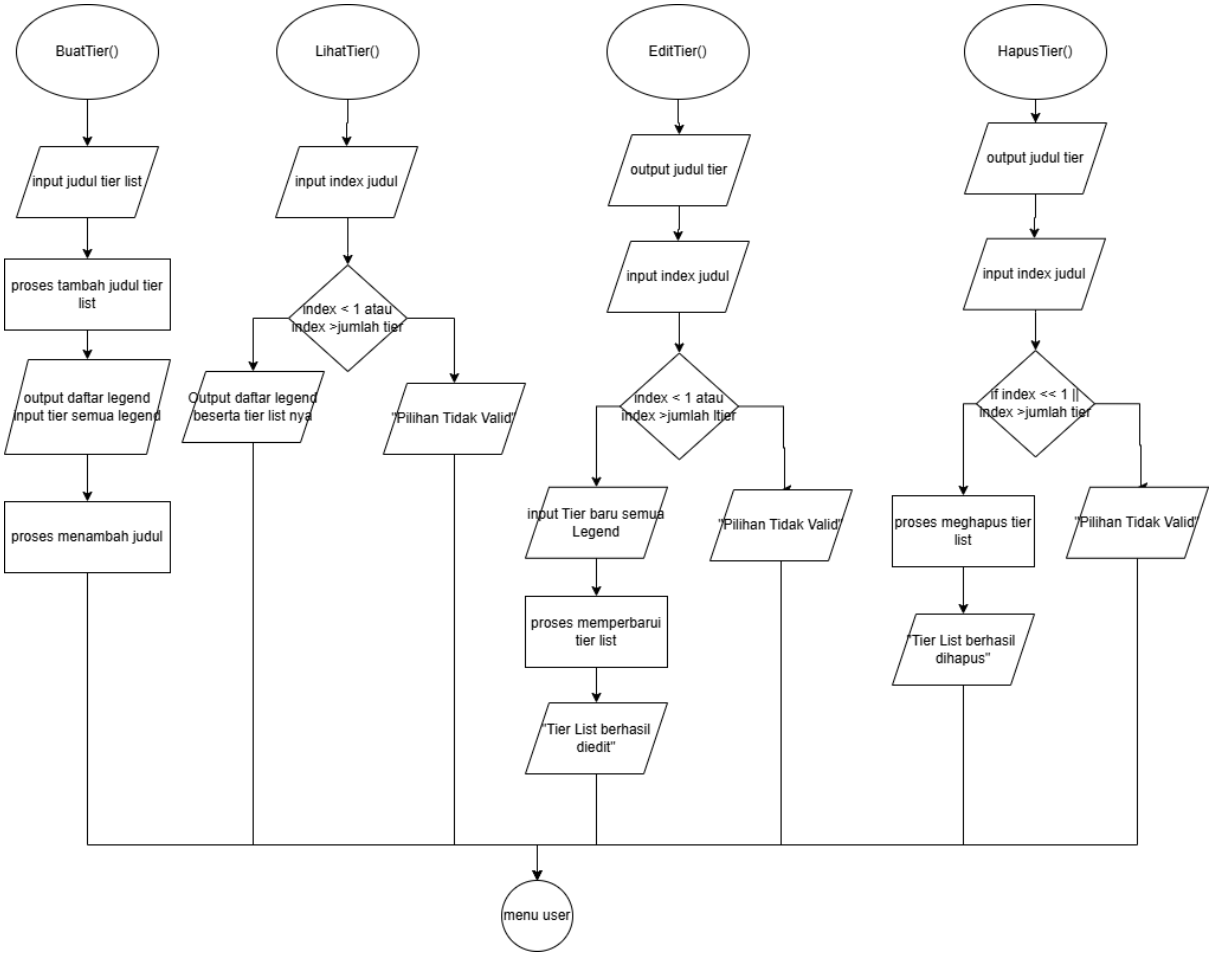

start
struct user (name, pw, role)
struct karakter (nama, class, weapon, pick rate, win rate)
struct tier list (judul, tier)
User dataUser[100];
karakter legend[30];
Tier_List tier_list[20];
int jumlahUser = 0;
int jumlahLegend = 0;
int jumlahtier = 0;
int percobaan = 3;
int pilihan;
Logout admin
Logout user
menu_login()
menu awal
"menu login
1. Login
2. Register
3. keluar"
input pilihan
pilihan == 1
pilihan == 2
pilihan == 3
loginUser()
RegisterUser()
"Terima kasih telah menggunakan program ini. Sampai jumpa!"
menu awal
end


menuAdmin()
menu admin
"Menu Admin
1. Tambah legend
2. Lihat daftar Legend
3. Edit Legend
4. Hapus Legend
5. Logout
input menu_admin
menu_admin == 1
menu_admin == 2
menu_admin == 3
menu admin == 4
menu admin == 5
TambahLegend()
DaftarLegend()
EditLegend()
HapusLegend()
Logout admin
menu admin


menuUser()
menu user
"Menu User
1. Lihat Daftar Legend
2. Buat Tier List
3. Lihat Tier List
4. Edit Tier List
5. Hapus Tier List
6. Logout
input menu_user
menu_user == 1
menu_user == 2
menu_user == 3
menu_user == 4
menu_user == 5
menu_user == 6
DaftarLegend()
BuatTier()
LihatTier()
EditTier()
HapusTier()
Logout user
menu user


loginUser()
input username
input password
check usn dan pw
"Login Gagal!"
percobaan - 1
if percobaan
== 0
"Program otomatis
selesai
dataUser[i].role
== admin
menuAdmin()
menuUser()
RegisterUser()
input username
input password
input role
proses
menambahkan user
"Register Berhasil"
menu
awal
end


TambahLegend()
input nama
class
weapon
pick rate
win rate
proses menambah karakter
"berhasil ditambah"
DaftarLegend()
jumlahLegend == 0
"belum ada Legend yang dapat ditampilkan"
output daftar legend
EditLegend()
ototput daftar legend
input index
index < 1 atau index >jumlah legend
"Pilihan Tidak Valid"
input nama
class
weapon
pick rate
win rate
proses memperbarui data karakter
"berhasil diedit"
HapusLegend()
otuput daftar legend
input index
index <1 atau index >jumlah legend
"Pilihan Tidak Valid"
proses menghapus legend
jumlahLegend--
"Legend berhasil dihapus"
menu admin


BuatTier()
input judul tier list
proses tambah judul tier list
output daftar legend input tier semua legend
proses menambah judul
LihatTier()
input index judul
index < 1 atau index >jumlah tier
Output daftar legend beserta tier list nya
"Pilihan Tidak Valid"
EditTier()
output judul tier
input index judul
index < 1 atau index >jumlah ltier
input Tier baru semua Legend
proses memperbarui tier list
"Tier List berhasil diedit"
"Pilihan Tidak Valid"
HapusTier()
output judul tier
input index judul
if index << 1 || index >jumlah tier
proses meghapus tier list
"Tier List berhasil dihapus"
"Pilihan Tidak Valid"
menu user

2. Deskripsi Singkat Program

Program ini bertemakan sistem data Karakter Game Apex Legend, di program ini terdapat dua role user saat login yakni admin dan user. Di menu login, kita bisa melakuakn registrasi terlebih dahulu jika belum memilki akun, setelah registrasi akan kembali ke halaman menu lagi lau bisa langsung login, tetapi ada batas untuk bisa melakukan login sebanyak 3 kali jika 3 kali salah maka program otomatis berhenti/selesai. Saat login program akan langsung mengetahui role apa akun tersebut.

Lanjut Kebagian CRUD, role admin Memiliki akses untuk melakukan CRUD pada karakter yang tersedia. Untuk user, user Bisa melakukan CRUD berupa Tier list dari karakter yang tersedia. Saat Membuat tier list user akan diminta memberi judul tierlist lalu user memberi tier pada karakter yang tersedia umtuk edit, user bisa mengedit ulang tier seluruh karakter, dan untuk hapus, user bisa menghapus tier list yang sudah dibaut.

3. Source Code

```
const int MAX_USER = 100;
const int MAX_LEGENDS = 30;
const int MAXLIST = 20;

struct User {
    string usn;
    string pw;
    string role;
};

struct karakter {
    string nama;
    string classs;
    string best_weapon;
    float pick_rate;
    float win_rate;
};
struct Tier_List {
    string judul;
    char tier[MAX_LEGENDS];
};


int main() {

    User dataUser[MAX_USER];
    karakter legend[MAX_LEGENDS];
    Tier_List tier_list[MAXLIST];
    int jumlahUser = 1;
    int jumlahLegend = 11;
    int jumlahtier = 0;;
    int percobaan = 3;
    int pilihan;
    dataUser[0] = {"admin", "admin123", "admin"};

    legend[0] = {"Alter", "Skirmisher", "SMG", 14.5, 52};
    legend[1] = {"Ash", "Skirmisher", "SMG", 0.4, 49.4};
    legend[2] = {"Bangalore", "Assault", "SMG", 7.4, 49.1};
    legend[3] = {"Fuse", "Assault", "SMG", 1.1, 48.8};
    legend[4] = {"Horizon", "Skirmisher", "AR", 3.6, 47.5};
    legend[5] = {"Loba", "Support", "AR", 2.8, 51.7};
    legend[6] = {"Newcastle", "Support", "SG", 0.6, 48.1};
    legend[7] = {"Seer", "Recon", "AR", 1.3, 48.7};
    legend[8] = {"valkyrie", "Recon", "AR", 9.2, 54.4};
    legend[9] = {"Wattson", "Controller", "SMG", 0.2, 51.7};
    legend[10] = {"Wraith", "Skirmisher", "SMG", 0.6, 47.5};
```

```cpp
void RegisterUser(User dataUser[], int &jumlahUser) {
    if (jumlahUser >= MAX_USER) {
    cout << "Maaf, kapasitas pengguna sudah penuh.\n";
    pause();
    return;
    }

    string usn, pw, role;
    cout << "Masukkan Username: ";
    cin >> usn;
    cout << "Masukkan Password: ";
    cin >> pw;
    cout << "Masukkan role (admin/user): ";
    cin >> role;

    if (role != "admin" && role != "user") {
        cout << "Role tidak valid. Harus 'admin' atau 'user'.\n";
        pause();
        return;
    }
    dataUser[jumlahUser].usn = usn;
    dataUser[jumlahUser].pw = pw;
    dataUser[jumlahUser].role = role;
    jumlahUser++;
    cout << "\nSelamat Registrasi berhasil! Silahkan Login\n";
    pause();
}

int LoginUser(User dataUser[], int jumlahUser){

    string usn, pw;

    cout << "Username: ";
    cin >> usn;

    cout << "Password: ";
    cin >> pw;

    for(int i = 0; i < jumlahUser; i++){
        if(dataUser[i].usn == usn && dataUser[i].pw == pw){
            return i;
        }
    }

    return -1;
}

void menu_login() {
    cout << "\n===== MENU LOGIN =====\n";
    cout << "1. Login\n";
    cout << "2. Register\n";
    cout << "3. Keluar\n";
    cout << "Pilih menu: ";
}

void menuAdmin() {
    cout << "\n===== MENU ADMIN =====\n";
    cout << "1. Tambah Legend\n";
    cout << "2. Lihat Daftar legend\n";
    cout << "3. Edit Legend\n";
    cout << "4. Hapus Legend\n";
    cout << "5. Logout\n";
    cout << "Pilih menu: ";
}

void menuUser() {
    cout << "\n===== MENU USER =====\n";
    cout << "1. Lihat Daftar Legend\n";
    cout << "2. Buat Tier List\n";
    cout << "3. Lihat Tier List\n";
    cout << "4. Edit Tier List\n";
    cout << "5. Hapus Tier List\n";
    cout << "6. Logout\n";
    cout << "Pilih menu: ";

}

```

```cpp
void TambahLegend(karakter legend[], int &jumlahLegend) {
    cout << "\nTambah Legend\n";
    cout << "Nama: ";
    cin >> legend[jumlahLegend].nama;
    cout << "Class: ";
    cin >> legend[jumlahLegend].classs;
    cout << "Best Weapon: ";
    cin >> legend[jumlahLegend].best_weapon;
    cout << "Pick Rate: ";
    cin >> legend[jumlahLegend].pick_rate;
    cout << "Win Rate: ";
    cin >> legend[jumlahLegend].win_rate;
    jumlahLegend++;
    cout << "Legend berhasil ditambahkan!\n";
    pause();
}

void DaftarLegend(karakter legend[], int &jumlahLegend) {
    cout << "\nLegend\n";
    if (jumlahLegend == 0) {
        cout << "Belum ada legend yang dapat ditampilkan.\n";
        pause();
    } else {
        cout << "\n Data Legend \n";
        cout << "|" << setw(12) << left << "Nama" << "|" << setw(12) << left << "Class" << "|" << setw(12) << left << "Best Weapon" << "|" << setw(18) << left << " Pick Rate" << "|" << setw(10) << left << " Win Rate |\n";
        cout << "==============================================================\n";
    for (int i = 0; i < jumlahLegend; i++) {
        cout << "| " << setw(18) << left << legend[i].nama << " | " << setw(10) << left << legend[i].classs << " | " << setw(10) << left << legend[i].best_weapon << " | " << setw(3) << left << legend[i].pick_rate << setw(5)<< "%" << " | " << setw(8) << left << legend[i].win_rate << "% |"  << "\n";
    }
    cout << "==============================================================\n";
    pause();
    }
}
```

```cpp
void EditLegend(karakter legend[], int &jumlahLegend) {
    cout << "\nEdit Legend\n";
    if (jumlahLegend == 0) {
        cout << "Belum ada legend yang dapat diedit.\n";
        pause();
    } else {
        cout << "\n Data Legend \n";
        cout << "|" << setw(12) << left << "Nama" << "|" << setw(12) << left << "Class" << "|" << setw(12) << left << "Best Weapon" << "|" << setw(18) << left << " Pick Rate" << "|" << setw(10) << left << " Win Rate |\n";
        cout << "==============================================================\n";
    for (int i = 0; i < jumlahLegend; i++) {
        cout << "| " << setw(18) << left << legend[i].nama << " | " << setw(10) << left << legend[i].classs << " | " << setw(10) << left << legend[i].best_weapon << " | " << setw(3) << left << legend[i].pick_rate << setw(5)<< "%" << " | " << setw(8) << left << legend[i].win_rate << "% |"  << "\n";
    }
    cout << "==============================================================\n";
        int index;
        cout << "\nPilih nomor legend yang ingin diedit: ";
        cin >> index;
        if (index < 1 || index > jumlahLegend) {
            cout << "Pilihan tidak valid.\n";
        } else {
            index--;
            cout << "\nEdit Legend: " << legend[index].nama << "\n";
            cout << "\nNama: ";
            cin >> legend[index].nama;
            cout << "Class: ";
            cin >> legend[index].classs;
            cout << "Best Weapon: ";
            cin >> legend[index].best_weapon;
            cout << "Pick Rate: ";
            cin >> legend[index].pick_rate;
            cout << "Win Rate: ";
            cin >> legend[index].win_rate;
            cout << "\nLegend berhasil diedit!\n";
            pause();
        }
    }
}

void HapusLegend(karakter legend[], int &jumlahLegend) {
    cout << "\nHapus Legend\n";
    if (jumlahLegend == 0) {
        cout << "Belum ada legend yang dapat dihapus\n";
        pause();
    } else {
        cout << "\n Data Legend \n";
        cout << "|" << setw(12) << left << "Nama" << "|" << setw(12) << left << "Class" << "|" << setw(12) << left << "Best Weapon" << "|" << setw(18) << left << " Pick Rate" << "|" << setw(10) << left << " Win Rate |\n";
        cout << "==============================================================\n";
    for (int i = 0; i < jumlahLegend; i++) {
        cout << "| " << setw(18) << left << legend[i].nama << " | " << setw(10) << left << legend[i].classs << " | " << setw(10) << left << legend[i].best_weapon << " | " << setw(3) << left << legend[i].pick_rate << setw(5)<< "%" << " | " << setw(8) << left << legend[i].win_rate << "% |"  << "\n";
    }
    cout << "==============================================================\n";
        int index;
        cout << "\nPilih nomor legend yang ingin dihapus: ";
        cin >> index;
        if (index < 1 || index > jumlahLegend) {
            cout << "Pilihan tidak valid.\n";
        } else {
            for (int i = index - 1; i < jumlahLegend - 1; i++) {
                legend[i] = legend[i + 1];
            }
            jumlahLegend--;
            cout << "\nLegend berhasil dihapus!\n";
            pause();
        }
    }
}
```

```
void BuatTier(Tier_List tier_list[], int &jumlahtier, karakter legend[], int jumlahLegend){
    cout << "\nBuat Tier List\n";
    cout << "Masukkan judul tier list: ";
    cin >> ws;
    getline(cin, tier_list[jumlahtier].judul);
    cout << "\nMasukkan tier setiap legend (S/A/B/C/D):\n";
    for ( int i = 0; i < jumlahLegend; i++) {
        cout << "Tier legend " << setw(5) << left << legend[i].nama << ": ";
        cin >> tier_list[jumlahtier].tier[i];
    }
    jumlahtier++;
    cout << "Tier list berhasil dibuat!\n";
    pause();
}

void LihatTIer(Tier_List tier_list[], int &jumlahtier, karakter legend[], int jumlahLegend){
    cout << "\nDaftar Tier List\n";
    if (jumlahtier == 0) {
        cout << "Belum ada tier list yang dapat ditampilkan.\n";
        pause();
        return;
    }else {
    for (int i = 0; i < jumlahtier; i++) {
        cout << i + 1 << ". " << tier_list[i].judul << "\n";
    }
    cout << endl;
    int index_list;
    cout << "\nPilih tier list yang ingin dilihat: ";
    cin >> index_list;
    if (index_list < 1 || index_list > jumlahtier) {
        cout << "Pilihan tidak valid.\n";
        pause();
        return;
    } else {
    index_list--;

    cout << "\n Tier List: " << tier_list[index_list].judul << "\n";
    for (int i = 0; i < jumlahLegend; i++) {
        cout << setw(10) << left << legend[i].nama << " = " << setw(2)<< tier_list[index_list].tier[i] << "\n";
            }
            pause();
        }
    }
}
```

```
void EditTier(Tier_List tier_list[], int &jumlahtier, karakter legend[], int jumlahLegend){
    cout << "\nDaftar Tier List.\n";
    if (jumlahtier == 0) {
        cout << "Belum ada tier list yang dapat diedit.\n";
        pause();
        return;
    } else {
    for (int i = 0; i < jumlahtier; i++) {
            cout << i + 1 << ". " << tier_list[i].judul <<"\n";
    }
    cout << endl;
    int index_edit;
    cout << "\nPilih tier list yang ingin diedit: ";
    cin >> index_edit;
    if (index_edit < 1 || index_edit > jumlahtier) {
        cout << "Pilihan tidak valid.\n";
        pause();
        return;
    } else {
    index_edit--;

    cout << "\nMasukkan tier baru setiap legend (S/A/B/C/D):\n";
    for (int i = 0; i < jumlahLegend; i++) {
        cout << "Masukkan Tier baru " << setw(5) << left << legend[i].nama << ": ";
        cin >> tier_list[index_edit].tier[i];
            }
        }
    }
    cout << "\nTier list berhasil diedit\n";
    pause();
}

void HapusTier(Tier_List tier_list[], int &jumlahtier, karakter legend[], int jumlahLegend){
    cout << "\nDaftar Tier List.\n";
    if (jumlahtier == 0) {
        cout << "Belum ada tier list yang dapat dihapus.\n";
        pause();
        return;
    } else {
    for (int i = 0; i < jumlahtier; i++) {
            cout << i + 1 << ". " << tier_list[i].judul << "\n";
    }
    int index_hapus;
    cout << "\nPilih tier list yang ingin dihapus: ";
    cin >> index_hapus;
    if (index_hapus < 1 || index_hapus > jumlahtier) {
        cout << "Pilihan tidak valid.\n";
        pause();
        return;
    } else {
    index_hapus--;

    for (int i = index_hapus; i < jumlahtier - 1; i++) {
        tier_list[i] = tier_list[i + 1];
            }
        }
    }
    jumlahtier--;
    cout << "\nTier list berhasil dihapus\n";
    pause();
}
```

```cpp
do {
    menu_login();
    cin >> pilihan;

    switch (pilihan) {
        case 1: {
            int indexUser = LoginUser(dataUser, jumlahUser);
            if(indexUser != -1){
                cout << "\nLogin berhasil! Selamat datang, "
                    << dataUser[indexUser].usn << " (" << dataUser[indexUser].role << ")\n";
                    pause();

                    if (dataUser[indexUser].role == "admin") {
                        int menu_admin;
                        do{
                            menuAdmin();
                            cin >> menu_admin;
                            switch (menu_admin) {
                                case 1:
                                    TambahLegend(legend, jumlahLegend);
                                break;

                                case 2:
                                    DaftarLegend(legend, jumlahLegend);
                                    break;

                                case 3:
                                    EditLegend(legend, jumlahLegend);
                                    break;

                                case 4:
                                    HapusLegend(legend, jumlahLegend);
                                    break;
                                case 5:
                                    logout();
                                    break;
                                default:
                                    cout << "Pilihan tidak valid. Silakan coba lagi.\n";
                                    pause();
                            }
                        } while (menu_admin != 5);
                    }
```

```
                        else if(dataUser[indexUser].role == "user") {
                            int menu_user;
                            do {
                                menuUser();
                                cin >> menu_user;
                                switch (menu_user) {
                                    case 1:
                                        DaftarLegend(legend, jumlahLegend);
                                        break;

                                    case 2:
                                        BuatTier(tier_list, jumlahtier, legend, jumlahLegend);
                                        break;

                                    case 3:
                                        LihatTIer(tier_list, jumlahtier, legend, jumlahLegend);
                                        break;

                                    case 4:
                                        EditTier(tier_list, jumlahtier, legend, jumlahLegend);
                                        break;

                                    case 5:
                                        HapusTier(tier_list, jumlahtier, legend, jumlahLegend);
                                        break;
                                    case 6:
                                        logout();
                                        break;

                                    default:
                                        cout << "Pilihan tidak valid. Silakan coba lagi\n";
                                        pause();
                                }
                            } while (menu_user != 6);
                        }
                        break;
                    }
                }

                if (indexUser == -1) {
                    percobaan--;
                    cout << "Login gagal! Sisa percobaan: " << percobaan << "\n";
                    pause();
                    if (percobaan == 0) {
                        cout << endl;
                        cout << "Anda telah kehabisan percobaan login. Program SELESAI.\n";
                        cout << endl;
                        exit(0);
                    }
                }
                break;
            case 2: {
                RegisterUser(dataUser, jumlahUser);
                break;
            }
            case 3:
                cout << endl;
                cout << "Terima kasih telah menggunakan program ini. Sampai jumpa!\n";
                cout << endl;
                break;
            default:
                cout << "Pilihan tidak valid. Silakan coba lagi.\n";
                pause();
        }
    } while (pilihan != 3);

    return 0;
}
```

4. Hasil Output

```
===== MENU LOGIN =====
1. Login
2. Register
3. Keluar
Pilih menu:
```

```
Pilih menu: 2
Masukkan Username: Cordova
Masukkan Password: 123
Masukkan role (admin/user): user
Selamat Registrasi berhasil! Silahkan Login
```

```
Pilih menu: 1

Masukkan Username: admin
Masukkan Password: admin123

Login berhasil! Selamat datang, admin (admin)
```

Gambar 4.1 Tampilan Login

```
===== MENU ADMIN =====
1. Tambah Legend
2. Lihat Daftar legend
3. Edit Legend
4. Hapus Legend
5. Logout
Pilih menu:
```

```
Pilih menu: 1

Tambah Legend
Nama: Octane
Class: Skimisher
Best Weapon: SMG
Pick Rate: 20.4
Win Rate: 3.7
Legend berhasil ditambahkan!
```

```
 Data Legend
|Nama        |Class       |Best Weapon | Pick Rate| Win Rate |
=================================================================
| Alter      | Skirmisher | SMG        | 14.5%     | 52% |
| Ash        | Skirmisher | SMG        | 0.4%     | 49.4% |
| Bangalore  | Assault    | SMG        | 7.4%     | 49.1% |
| Fuse       | Assault    | SMG        | 1.1%     | 48.8% |
| Horizon    | Skirmisher | AR         | 3.6%     | 47.5% |
| Loba       | Support    | AR         | 2.8%     | 51.7% |
| Newcastle  | Support    | SG         | 0.6%     | 48.1% |
| Seer       | Recon      | AR         | 1.3%     | 48.7% |
| valkyrie   | Recon      | AR         | 9.2%     | 54.4% |
| Wattson    | Controller | SMG        | 0.2%     | 51.7% |
| Wraith     | Skirmisher | SMG        | 0.6%     | 47.5% |
| Octane     | Skimisher  | SMG        | 20.4%     | 3.7% |
=================================================================
```

```
 Data Legend
|Nama         |Class        |Best Weapon | Pick Rate| Win Rate |
==================================================================
| Alter       | Skirmisher | SMG         | 14.5%     | 52% |
| Ash         | Skirmisher | SMG         | 0.4%     | 49.4% |
| Bangalore   | Assault    | SMG         | 7.4%     | 49.1% |
| Fuse        | Assault    | SMG         | 1.1%     | 48.8% |
| Horizon     | Skirmisher | AR          | 3.6%     | 47.5% |
| Loba        | Support    | AR          | 2.8%     | 51.7% |
| Newcastle   | Support    | SG          | 0.6%     | 48.1% |
| Seer        | Recon      | AR          | 1.3%     | 48.7% |
| valkyrie    | Recon      | AR          | 9.2%     | 54.4% |
| Wattson     | Controller | SMG         | 0.2%     | 51.7% |
| Wraith      | Skirmisher | SMG         | 0.6%     | 47.5% |
| Octane      | Skimisher  | SMG         | 20.4%     | 3.7% |
==================================================================

Pilih nomor legend yang ingin diedit: 12

Edit Legend: Octane

Nama: Octane
Class: Skimisher
Best Weapon: SG
Pick Rate: 30.2
Win Rate: 51.4

Legend berhasil diedit!
```

```
 Data Legend
|Nama        |Class       |Best Weapon | Pick Rate| Win Rate |
=================================================================
| Alter      | Skirmisher | SMG        | 14.5%     | 52% |
| Ash        | Skirmisher | SMG        | 0.4%     | 49.4% |
| Bangalore  | Assault    | SMG        | 7.4%     | 49.1% |
| Fuse       | Assault    | SMG        | 1.1%     | 48.8% |
| Horizon    | Skirmisher | AR         | 3.6%     | 47.5% |
| Loba       | Support    | AR         | 2.8%     | 51.7% |
| Newcastle  | Support    | SG         | 0.6%     | 48.1% |
| Seer       | Recon      | AR         | 1.3%     | 48.7% |
| valkyrie   | Recon      | AR         | 9.2%     | 54.4% |
| Wattson    | Controller | SMG        | 0.2%     | 51.7% |
| Wraith     | Skirmisher | SMG        | 0.6%     | 47.5% |
| Octane     | Skimisher  | SG         | 30.2%     | 51.4% |
=================================================================

Pilih nomor legend yang ingin dihapus: 1

Legend berhasil dihapus!
```

```
===== MENU ADMIN =====
1. Tambah Legend
2. Lihat Daftar legend
3. Edit Legend
4. Hapus Legend
5. Logout
Pilih menu: 5
Logout berhasil. Kembali ke menu utama.
```

Gambar 4.2 Tampilan Menu dan Fitur Admin

```
Masukkan Username: Cordova
Masukkan Password: 123

Login berhasil! Selamat datang, Cordova (user)

===== MENU USER =====
1. Lihat Daftar Legend
2. Buat Tier List
3. Lihat Tier List
4. Edit Tier List
5. Hapus Tier List
6. Logout
Pilih menu:
```

```
Buat Tier List
Masukkan judul tier list: Tier list Ultimate Legend

Masukkan tier setiap legend (S/A/B/C/D):
Tier legend Ash  : S
Tier legend Bangalore: S
Tier legend Fuse : A
Tier legend Horizon: B
Tier legend Loba : B
Tier legend Newcastle: A
Tier legend Seer : C
Tier legend valkyrie: S
Tier legend Wattson: D
Tier legend Wraith: B
Tier legend Octane: A
Tier list berhasil dibuat!
```

```
Pilih menu: 3

Lihat Tier List
1. Tier list Ultimate Legend

Pilih tier list yang ingin dilihat: 1

 Tier List: Tier list Ultimate Legend
Ash         = S
Bangalore   = S
Fuse        = A
Horizon     = B
Loba        = B
Newcastle   = A
Seer        = C
valkyrie    = S
Wattson     = D
Wraith      = B
Octane      = A
```

```
Pilih menu: 4

Daftar Tier List.
1. Tier list Ultimate Legend

Pilih tier list yang ingin diedit: 1

Masukkan tier baru setiap legend (S/A/B/C/D):
Masukkan tier Baru Ash:    C
Masukkan tier Baru Bangalore:    A
Masukkan tier Baru Fuse:    S
Masukkan tier Baru Horizon:    A
Masukkan tier Baru Loba:    A
Masukkan tier Baru Newcastle:    B
Masukkan tier Baru Seer:    C
Masukkan tier Baru valkyrie:    S
Masukkan tier Baru Wattson:    B
Masukkan tier Baru Wraith:    A
Masukkan tier Baru Octane:    B

Tier list berhasil diedit
```

```
Pilih menu: 5

Daftar Tier List.
1. Tier list Ultimate Legend

Pilih tier list yang ingin dihapus: 1

Tier list berhasil dihapus
```

Gambar 4.3 Tampilan Menu dan Fitur User

```
===== MENU LOGIN =====
1. Login
2. Register
3. Keluar
Pilih menu: 3

Terima kasih telah menggunakan program ini. Sampai jumpa!
```

Gambar 4.4 Tampilan Keluar Program

5. Langkah-langkah GIT

5.1 GIT Add

berfungsi untuk menambahkan file atau perubahan ke staging area, sebelum disimpan permanen

```
PS D:\Tugas Kuliah\Tugas Praktikum\praktikum-apl> git add post-test
```

5.2 GIT Commit

befungsi untuk menyimpan perubahan yang sudah ada di staging area ke dalam repository Git beserta pesan (message) sebagai catatan perubahan

```
PS D:\Tugas Kuliah\Tugas Praktikum\praktikum-apl> git commit -m "fix minor"
[main c6988b6] fix minor
 2 files changed, 19 insertions(+), 19 deletions(-)
```

5.3 GIT Push

berfungsi untuk mengirim atau (upload) commit dari repository local ke repository Github

```
PS D:\Tugas Kuliah\Tugas Praktikum\praktikum-apl> git push
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.44 KiB | 133.00 KiB/s, done.
Total 6 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 5 local objects.
To https://github.com/Cordova88/praktikum-apl.git
   e1244f6..c6988b6  main -> main
```